Transforming NASA Earth Observation Data into Actionable Farming Decisions

AgriPulse

NASA SPACE APPS CHALLENGE
EARTH OBSERVATION · SUSTAINABLE AGRICULTURE



Gemini Notebook

A passionate student team democratizing space technology for sustainable agriculture



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Scientific & Technical Mentor

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Farmers need better signals sooner to prevent resource waste and crop failure

The Reality in the Field

- Guesswork-based irrigation can waste scarce freshwater.
- Late recognition of water stress and disease can threaten yields.

The Disconnected Data

- Valuable NASA satellite and climate data exists but remains locked away.
- These signals are traditionally too hard for everyday farmers to access and interpret without technical expertise.

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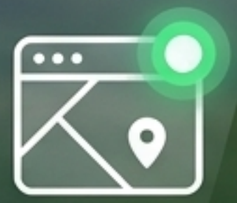
AgriPulse is a user-friendly web platform translating Earth observation into farming decisions



Automated data fetching

Less searching.
More time acting.

Meet AgriPulse

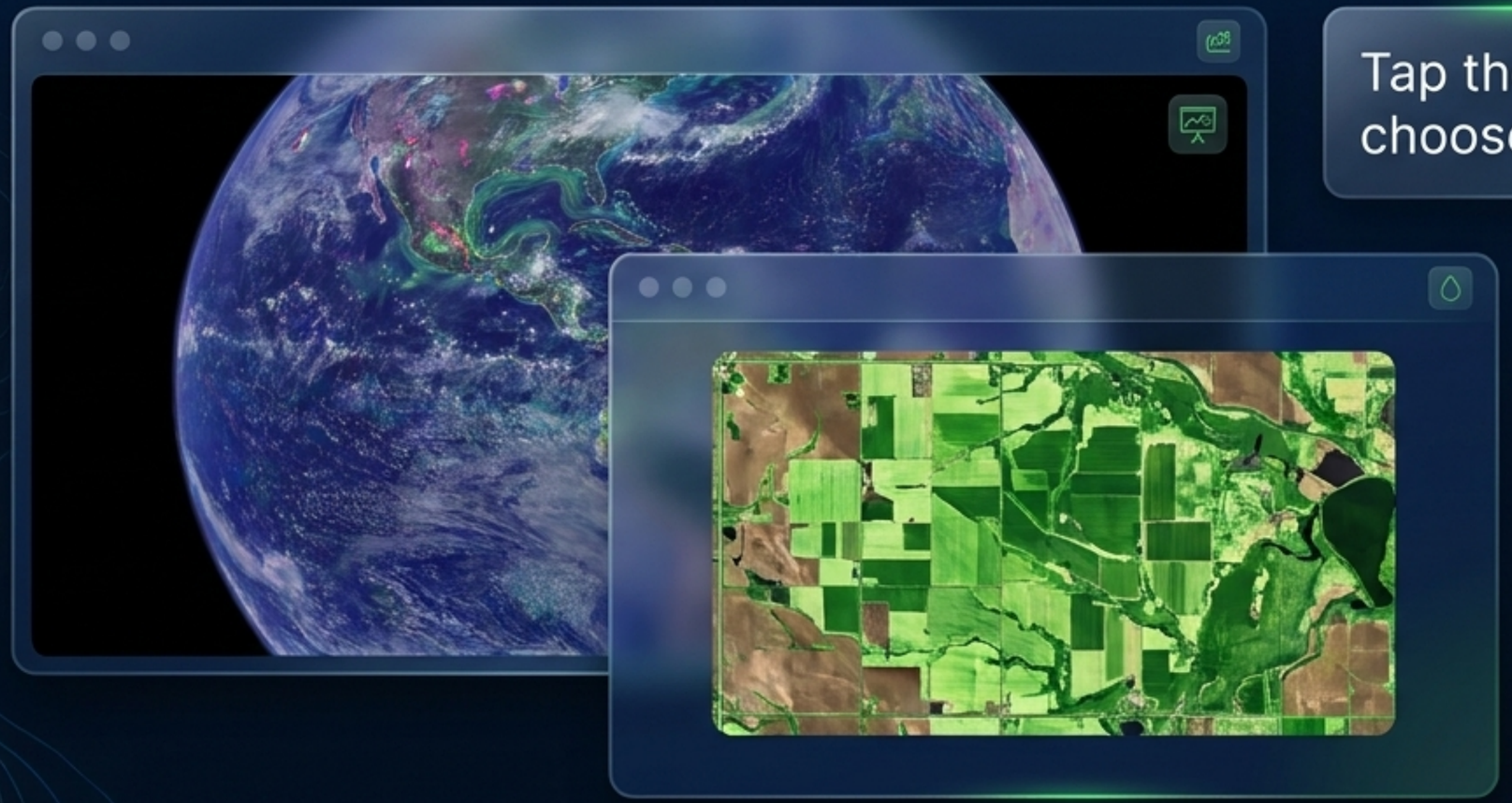


Visual maps and timely alerts

Signals shaped for the field.

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Interactive mapping seamlessly connects global meteorological data to local fields



Tap the map to choose your field.

Powered by Landsat and MODIS imagery, plus NASA POWER meteorological data.

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The orbital pipeline automates the journey from NASA data to irrigation decisions



Mapping Field Boundaries

Defining the exact geographic area of interest.



Ingesting NDVI + Moisture Indicators

Pulling targeted Earth observation and weather data.



Pushing Automated Irrigation Alerts

Delivering timely, actionable signals directly to the user.

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Precision insights deliver measurable impact across water, economics, and access

Water

Target up to 30–40% less irrigation waste

(Note: A project target requiring field validation, not a measured result)

Farm Economics

Potential to protect yield and significantly lower operating costs.

Access

100% Browser-based, requiring absolutely no expensive field hardware.

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Bridging the gap between the stars and the soil



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